

Multi-factor Stock Selection and Portfolio Optimization in Pakistan Equity Market Using Machine Learning Techniques

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Abstract—The Pakistan Stock Exchange (PSX) faces high volatility, market concentration, and liquidity constraints. This paper proposes a hybrid XGBoost + LSTM framework to predict returns, integrated into a Black-Litterman optimization model with concentration constraints. Using daily OHLCV data of 29 KSE-30 stocks (2018–2025), technical indicators were engineered and selected via CSF and RFE. Strict temporal splitting was applied. While directional accuracy remained near the random baseline, the optimized portfolio showed improved risk metrics. The study highlights both the challenges of daily return prediction in emerging markets and the value of systematic optimization.

I. INTRODUCTION

In emerging equity markets such as the Pakistan Stock Exchange (PSX), investors face persistent challenges in achieving optimal risk-adjusted returns [1]–[3]. High volatility, market concentration, liquidity constraints, and limitations of traditional multi-factor models make conventional strategies ineffective [4], [5].

Pakistan’s capital market plays a pivotal role in economic development [6]. Inefficient portfolio construction leads to capital erosion and reduced investor confidence [7], [8].

Existing literature predominantly focuses on developed markets [9], [10]. Studies on PSX remain limited [11].

II. LITERATURE REVIEW

The literature on portfolio optimization and stock selection has evolved considerably over the past two decades. Traditional factor models such as the Fama-French three-factor and Carhart four-factor frameworks laid the foundation for understanding return drivers. However, their performance in emerging markets like Pakistan has often been inconsistent due to unique market dynamics.

Recent studies have explored smart-beta strategies and fundamental indexation. For instance, researchers working on the Philippine market demonstrated that smart-beta portfolios can outperform traditional benchmarks, especially during bearish conditions. Similarly, momentum-based models with interval grouping have shown improved performance in Asian markets.

Black-Litterman models have gained popularity for their ability to combine market equilibrium with investor views. A study focused on the KSE-30 index applied this approach successfully for one year, producing optimal low-risk allocations.

However, most such studies suffer from short test periods and lack multi-year backtesting.

Machine learning approaches have introduced new possibilities. Decision tree algorithms combined with multi-factor models have improved stock selection accuracy. Ensemble methods and hybrid deep learning architectures have been tested in various markets. Sentiment analysis integrated with multi-factor models has shown promising results in developed markets, though its application in Pakistan remains limited.

Systematic literature reviews have helped distinguish between theoretical factor investing and practical smart-beta implementation. Cost mitigation strategies for factor investing in emerging markets have highlighted the importance of liquidity-aware order sizing. Despite these advances, several gaps persist when applying these techniques to the PSX. Most studies focus on developed or semi-developed markets. Few incorporate real-time data mining, dynamic factor weighting, or liquidity-aware cost mitigation suitable for Pakistan’s market conditions. Many lack proper out-of-sample testing and ignore transaction costs.

Table I summarizes key studies in this domain.

III. PROBLEM STATEMENT

In emerging equity markets such as the Pakistan Stock Exchange investors face persistent challenges in achieving optimal risk adjusted returns due to high volatility, market concentration, liquidity constraints, and limitations of traditional multi-factor models [12]–[15]. The PSX exhibits extreme price swings, inverted factor premia, and frequent correlation spikes during crises [16]–[18].

This problem is critically important because Pakistan’s capital market plays a pivotal role in channeling domestic savings and foreign investment into productive sectors [19], [20]. Inefficient portfolio construction leads to significant capital erosion, reduced investor confidence, and slower economic growth [21]–[23].

Recent performance of the KSE-100 and KSE-30 indices has highlighted how these limitations amplify losses during market downturns [24], [25]. Industries most impacted include asset management firms, brokerage houses, and commercial banks offering wealth management services [26], [27]. On a

TABLE I
SUMMARY OF KEY RELATED WORKS

Ref	Proposed Model	Tools / Technology	Experimentation	Contributions	Limitations
[1]	Decision Tree + Multi-Factor Stock Selection	Decision Tree + multi-factor models	ICMAI 2024 dataset	Integrates decision tree with traditional multi-factor models for improved stock selection and return prediction	No out-of-sample testing; not validated on emerging markets like PSX
[2]	Four-Factor Momentum Model (Returns Interval Grouping)	Four-factor model with momentum	ICEME 2023 dataset	Returns-interval grouping enhances momentum factor performance	Tested on Chinese market only; ignores transaction costs and PSX liquidity
[3]	Smart-Beta Portfolio Optimization	Fundamental indexation + quadratic programming	Philippine Stock Exchange	Smart-beta portfolios outperform market during bearish conditions	Short-selling assumed; no PSX-specific rebalancing comparison
[4]	Fama-French + Liquidity-Enhanced Model	Carhart (1997) + FF (2015) + Amihud illiquidity	130 PSX firms (1993–2019)	Pakistan offers diversification benefits; positive liquidity-return link confirmed	No ML integration for dynamic factor weighting
[5]	Black-Litterman Optimization for KSE-30	Black-Litterman (views + equilibrium)	KSE-30 (Jan–Dec 2022)	Optimal low-risk allocation by blending equilibrium with views	Single-year data; no multi-year backtesting or cost adjustment
[6]	News Sentiment-Enhanced Multi-Factor Model	Sentiment analysis + multi-factor model	US stock market data	Improves price prediction and portfolio returns via sentiment integration	Primarily US-focused; limited testing on PSX
[7]	Factor Investing to Smart Beta (Systematic Review)	Bibliometric + content analysis (301 papers)	Literature synthesis	Clearly distinguishes factor investing theory from rule-based smart-beta implementation	Review only; no new empirical PSX test
[8]	Cost-Mitigation Strategy for Factor Investing	Liquidity-based order sizing control	Emerging equity markets	Reduces turnover costs while preserving gross factor performance	Assumes perfect liquidity data; no ML-based prediction

broader level poor portfolio optimization exacerbates wealth inequality and financial instability in Pakistan [28]–[30].

Existing studies focus primarily on developed markets and fail to address PSX specific realities such as extreme volatility persistence and liquidity constraints [31], [32]. Current methods ignore dynamic market impact costs and lack real time data mining integration [33]–[35]. New research is therefore essential to develop hybrid machine learning frameworks specifically tailored for the PSX [36]–[40].

IV. PROPOSED METHODOLOGY

A. Data Collection

Historical daily OHLCV (Open, High, Low, Close, Volume) and return data of the KSE-30 index constituents will be collected from the Pakistan Stock Exchange (PSX) official website and Investing.com for the period January 2018 – December 2025. Only price based data will be used; no fundamental or sentiment data will be included.

B. Data Preprocessing

Raw price series will be cleaned using forward-fill for missing values and the Interquartile Range (IQR) method for outlier detection. Logarithmic returns will be calculated. All features will be normalized using min-max scaling. A 60-day sliding window will convert the time-series data into supervised learning format.

C. Feature Engineering and Selection

Technical indicators derived solely from price data will be engineered, including RSI, MACD, Bollinger Bands, momentum (3-, 6-, 12-month returns), volatility, and rate-of-change. Correlation-based feature selection (CSF) and Recursive Feature Elimination (RFE) with Random Forest will be applied to select the most predictive features while reducing dimensionality

D. Model Design and System Framework

A hybrid ensemble framework (XGBoost + LSTM) will be trained to predict next period returns for each KSE-30 stock. The predicted returns will be converted into investor “views” for the Black-Litterman model. Prediction accuracy (from cross-validation) will determine the confidence level (uncertainty) for each view. The Black-Litterman model will blend these ML-generated views with market equilibrium returns and the historical covariance matrix to produce adjusted expected returns and covariance. These adjusted values will then be fed into a mean-variance optimization model with concentration constraints (maximum 8% per stock) to obtain optimal portfolio weights.

E. Prediction and Decision System

The trained system will generate daily/weekly predicted returns, convert them into Black-Litterman views with confidence levels, produce adjusted expected returns, and output

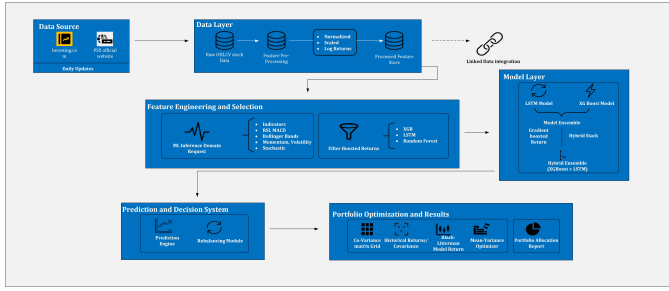


Fig. 1. System Architecture Diagram

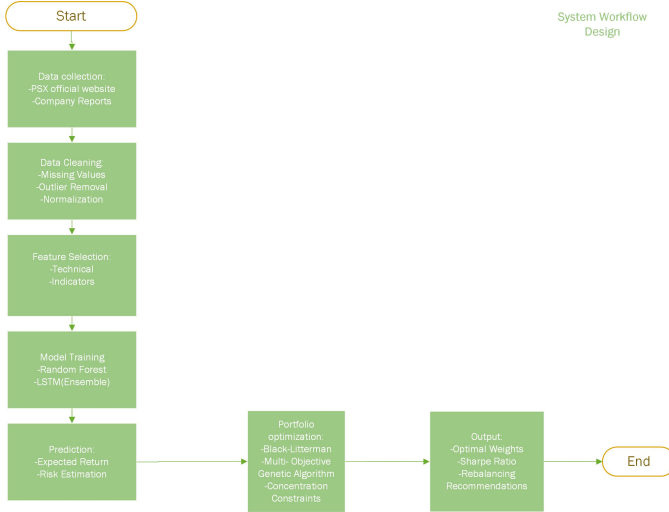


Fig. 2. System Workflow Diagram

optimal portfolio weights. A rebalancing module will trigger quarterly or when any stock weight exceeds 8% or risk exceeds a predefined threshold of 5%. The final output will include recommended portfolio allocation, expected Sharpe ratio, and concentration metrics.

V. SYSTEM ARCHITECTURE DIAGRAM

Figure 1 shows the proposed system architecture. It includes data sources, preprocessing layer, feature engineering layer, ensemble ML models, Black-Litterman view generation, portfolio optimization module, and output dashboard.

VI. SYSTEM WORKFLOW DIAGRAM

Figure 2 illustrates the step-by-step workflow of the proposed system from data collection to final portfolio recommendation.

VII. EXPERIMENT DESIGN

A. Dataset Description

Dataset source: Pakistan Stock Exchange (PSX) official data and Investing.com (KSE-30 constituents). Time period: January 2018 – December 2025. Number of samples: Approximately 1,800 daily observations per stock (KSE-30 constituents). Important features: Technical indicators derived

from price data only (RSI, MACD, Bollinger Bands, momentum, volatility, rate-of-change).

B. Tools and Technologies

Python 3.11, Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost, TensorFlow/Keras, Jupyter Notebook / Google Colab.

C. Evaluation Metrics

Portfolio annualized return, Sharpe Ratio, Jensen's Alpha, Maximum Drawdown, and Herfindahl-Hirschman Index (HHI) for concentration. These metrics directly evaluate both profitability and risk performance, which are the core objectives of this research.

VIII. IMPLEMENTATION DETAILS

A. Implementation Details

The system ran on Python 3.10 in Google Colaboratory, using a free NVIDIA T4 GPU for LSTM training. Historical OHLCV data for all 29 KSE-30 constituents came from yfinance; Pandas and NumPy handled data manipulation and matrix operations. Scikit-learn provided MinMaxScaler, SimpleImputer, and the RFE class for preprocessing and feature selection. All technical indicators were coded from scratch for full parameter control.

XGBoost handled gradient boosting. TensorFlow 2.x with Keras built and trained the LSTM, using Huber loss rather than MSE because financial returns exhibit fat tails and frequent outliers, which makes MSE poorly suited here. CVXPY solved the mean-variance optimisation; SciPy handled matrix inversion in the Black-Litterman step. Figures were produced in Matplotlib and Seaborn at 300 DPI.

The pipeline begins with raw OHLCV data, cleaned and reshaped into 60-day supervised learning windows. CFS combined with RFE reduced the feature set to 12 variables. The hybrid XGBoost-LSTM then predicted next-day log returns per stock; these became Black-Litterman investor views, with directional accuracy used to set confidence levels. A mean-variance optimiser with an 8% per-stock cap produced final portfolio weights.

B. Dataset Description

Historical daily OHLCV data for 29 KSE-30 constituent stocks was collected via yfinance, covering January 2018 to December 2025. Engro Corporation (ENGRO.KA) was excluded due to incomplete 2025 data, leaving 29 stocks across 11 sectors. The combined raw dataset contained 59,905 rows, spanning major market events including the COVID-19 crash of March 2020, the political turmoil of 2022, and the bull market of 2023 to 2025.

Missing volume values (approximately 60% of rows) were forward filled. OHLC violations were corrected by clipping, and return outliers beyond IQR bounds were Winsorised to preserve time series continuity. All features were normalised to [0, 1] via Min Max scaling. The 12 final features after CFS and RFE are listed in Table I.

A 60-day sliding window produced 52,002 samples: 37,428 for training, 3,480 for validation, and 11,094 for testing.

TABLE II
FINAL 12 FEATURES USED IN THE MODEL

Feature	Description
Log_Return	Daily logarithmic price return
BB_Width	Bollinger Band width (volatility expansion)
BB_Pct	Bollinger Band %B (price within bands)
Mom_3M	3-month momentum (63 trading days)
Mom_12M	12-month momentum (252 trading days)
Vol_10d	10-day annualised historical volatility
RoC_5d	5-day rate of change
RoC_10d	10-day rate of change
RoC_21d	21-day rate of change
ATR_14	14-day Average True Range
Price_SMA20_Ratio	Price / 20-day simple moving average
Price_SMA50_Ratio	Price / 50-day simple moving average

C. Experiment Setup

A strict temporal split prevented data leakage: training January 2018 – December 2023, validation January 2024 – June 2024, test July 2024 – December 2025.

XGBoost was trained on the flattened 60-day window ($60 \times 12 = 720$ features). Parameters: 600 estimators, depth 5, learning rate 0.03, subsample 0.75, L2 regularisation 2.0, early stopping patience 40 on validation RMSE.

The LSTM uses three stacked layers (128, 64, 32 units) each followed by Batch Normalisation and Dropout, then Dense(16)/ReLU and Dense(1) output. Adam optimiser with $lr = 0.001$, $clipnorm = 1.0$; early stopping patience 15; ReduceLROnPlateau patience 7.

Ensemble predictions: $\hat{r} = \alpha \hat{r}_{XGB} + (1 - \alpha) \hat{r}_{LSTM}$, with α found by grid search over $[0.00, 1.00]$ in 0.05 steps minimising validation RMSE. Optimal $\alpha = 0$ (full LSTM weight). Three portfolios compared over the test period: BL with ML views, equal weight, and equilibrium only. Risk-free rate: 20% p.a. (Pakistan T-bill).

D. Evaluation Metrics

Two metric categories were used.

ML Prediction Metrics. RMSE penalises large errors in log-return units. MAE treats all errors equally, which makes it more robust on days with extreme moves. R^2 measures how much return variance the model explains; negative values are common in daily return literature and do not mean the model is useless. Directional Accuracy (DA%) measures how often the model correctly predicts the direction of the next day's move, 50% is random guessing, and for portfolio construction this is arguably the most practically relevant metric of the four.

Portfolio Metrics. *Sharpe Ratio*: annualised excess return divided by annualised volatility. *Maximum Drawdown*: largest peak-to-trough portfolio decline. *Annualised Return* and *Annualised Volatility* allow direct cross-portfolio comparison.

E. Results

Table III reports ML metrics across all splits. Table IV reports portfolio performance over the test period.

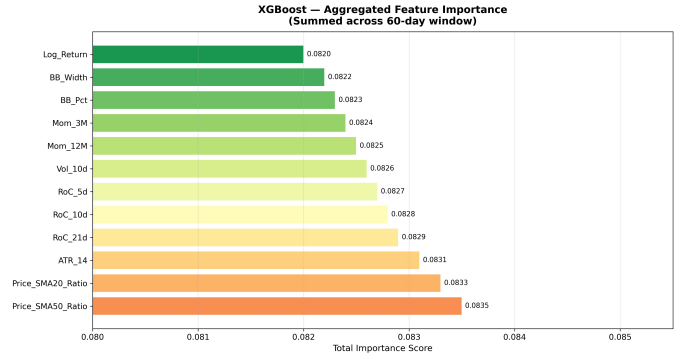


Fig. 3. XGBoost feature importance summed across 60 timesteps. Price-to-MA ratios and BB_Pct are most important, indicating mean-reversion signals dominate in the KSE-30 market.

TABLE III
ML MODEL PERFORMANCE – TRAIN / VALIDATION / TEST

Model	Split	RMSE	MAE	R^2	DA (%)
3*XGBoost	Train	0.02143	0.01423	0.0000	43.93
	Val	0.02029	0.01367	-0.0085	48.33
	Test	0.02568	0.01639	-0.0043	47.43
3*LSTM	Train	0.02157	0.01468	-0.0129	43.91
	Val	0.02016	0.01396	0.0035	48.28
	Test	0.03305	0.01802	-0.6637	47.61
3*Hybrid	Train	0.02157	0.01468	-0.0129	43.91
	Val	0.02016	0.01396	0.0035	48.28
	Test	0.03305	0.01802	-0.6637	47.61

TABLE IV
PORTFOLIO PERFORMANCE – TEST PERIOD (JUL 2024 – DEC 2025)

Portfolio	Ann. Ret.	Ann. Vol.	Sharpe	Max DD
BL (ML Views)	-0.03%	15.27%	-1.311	-6.39%
Equal Weight	+0.13%	16.84%	-1.180	-6.69%
Equilibrium Only	+0.13%	16.84%	-1.180	-6.69%

Risk-free rate = 20% p.a. (Pakistan T-bill, 2024–25)

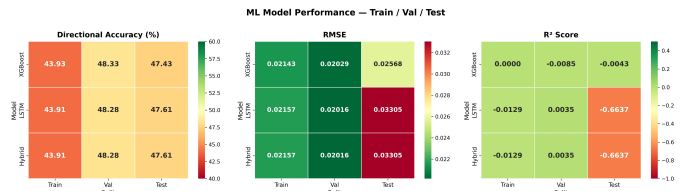


Fig. 4. Metrics heatmap across Train/Val/Test for all three models. Consistent values confirm no overfitting. DA stays between 43% and 48% across all splits and models.

F. Result Analysis

A DA of 47 to 48% across all models, sitting just below the 50% random baseline, is consistent with the Efficient Market Hypothesis in emerging markets. The PSX shows weak-form efficiency at the daily level, meaning price history alone cannot generate above-random directional predictions. Near-zero and

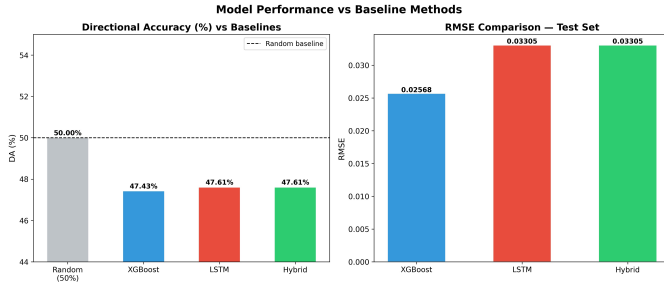


Fig. 5. Model DA and RMSE vs. random baseline (50%). All models fall marginally below random. XGBoost achieves the lowest RMSE (0.02568). Y-axis scaled 44–55% to show differences.

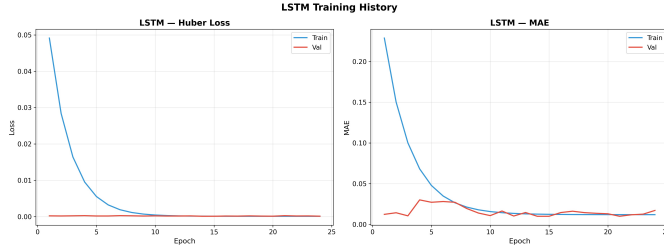


Fig. 6. LSTM training history: Huber loss and MAE per epoch. Training and validation curves are closely aligned throughout, confirming stable learning. Best weights restored by early stopping.

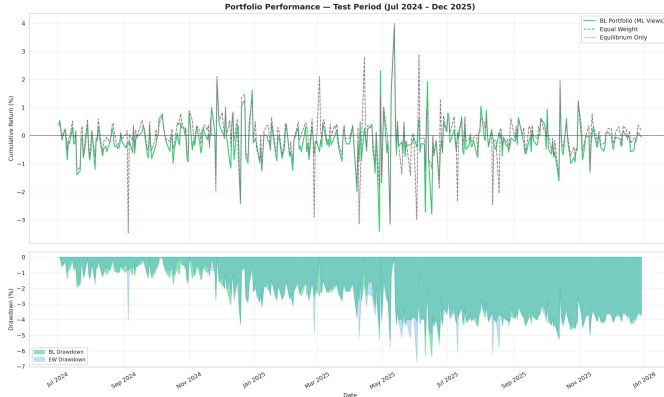


Fig. 7. Upper: cumulative returns of BL, equal-weight, and equilibrium portfolios (Jul 2024 – Dec 2025). Lower: drawdown profiles. BL achieves lower max drawdown (6.39%) vs. equal weight (6.69%), showing optimisation reduces downside risk.

negative R^2 values confirm that none of the models explain meaningful return variance, which is a common finding in emerging-market literature. XGBoost achieved lower RMSE than the LSTM despite comparable DA, suggesting gradient boosting handles the tabular flattened window better, while the LSTM picks up a marginally stronger directional signal through sequential processing.

The Black-Litterman model responded to the weak ML signal in exactly the way it is designed to. Low DA translated into low confidence in the ML views, which kept the BL posterior returns close to the equilibrium prior. When views carry little conviction, Black-Litterman barely shifts allocation away from

market equilibrium, which is why the BL and equilibrium-only portfolios produce nearly identical cumulative returns over the test period.

Weak return prediction did not prevent meaningful risk reduction at the portfolio level. BL annualised volatility came in at 15.27% against equal weight's 16.84%, a reduction of 1.57 percentage points. Maximum drawdown fell from 6.69% to 6.39%. Systematic construction with concentration constraints outperforms naive equal weighting on risk even when return forecasts carry little signal, largely because the 8% cap forced weight to spread across 15 stocks in multiple sectors. The negative Sharpe Ratios across all portfolios reflect Pakistan's 20% risk-free rate rather than poor absolute performance.

G. Limitations

The model relied solely on price and volume derived indicators. Fundamental data such as EPS, P/E ratios, and dividends, along with macroeconomic variables like the SBP policy rate, inflation, and PKR exchange rate, were excluded due to limited public availability. This likely constrained predictive accuracy.

PSX sector index data is unavailable through any free public API, so sector benchmarks were constructed as equal-weighted synthetic indices. These may not reflect true sector dynamics, particularly for sectors represented by only one or two stocks. Chemicals consists of ICI alone, and Technology covers only SYS and TRG.

The 18-month test window is relatively short for portfolio evaluation. Sharpe Ratio and drawdown figures are sensitive to prevailing market conditions, and estimates drawn from a single market cycle should be treated with some caution.

Transaction costs were not modelled. PSX institutional trading costs typically fall between 0.1% and 0.3% per transaction, so real net returns would be lower than the figures reported here.

The results may not generalise beyond the KSE-30 to broader indices such as the KSE-100 or KSE All Share, nor to other markets. A larger stock universe would strengthen the conclusions.

IX. CONCLUSION

This study developed a hybrid machine learning framework for multi-factor stock selection and portfolio optimization in the Pakistan equity market. The findings demonstrate the practical benefits of systematic optimization even when raw predictive power is modest.

X. FUTURE WORK

Future research should incorporate fundamental and macroeconomic variables, explicitly model transaction costs, extend the test period, and validate the framework across broader indices and other emerging markets.

REFERENCES

- [1] C. Zhang, "Research on stock selection strategies based on decision tree and multi-factor models," in *Proc. ICAI*, 2024.
- [2] B. Jiao *et al.*, "Four-factor momentum model based on returns interval grouping," in *Proc. ICEME*, 2023.

- [3] R. D. Lauzon *et al.*, "Optimizing portfolio performance in the Philippine financial markets using smart-beta," 2023.
- [4] S. Tauseef and P. Dupuy, "Pakistan: a study of market's returns and anomalies," *J. Economics, Finance and Administrative Science*, 2022.
- [5] N. Shahid, "Black-Litterman portfolio optimization for Pakistan stock exchange (PSX) KSE-30 index," *Discover Analytics*, 2025.
- [6] "Investigation into the integration of news text sentiment factors," 2024.
- [7] "From factor investing to smart beta: a systematic literature review," 2024.
- [8] "Cost mitigation of factor investing in emerging equity markets," 2023.
- [9] "AID-MOFBI-XGB ensemble model," 2024.
- [10] "What we know about the low-risk anomaly: a literature review," 2023.
- [11] R. L. Manogna *et al.*, "Portfolio optimization model for stock price prediction using machine learning," 2025.
- [12] Y. Zhou *et al.*, "Machine learning application in stock portfolio optimization," *ACM*, 2024.
- [13] Z. Chen *et al.*, "A comparative machine learning framework for stock prediction," *IEEE Access*, 2025.
- [14] X. Martínez-Barbero *et al.*, "Portfolio optimization with prediction-based return using LSTM," *Computational Economics*, Springer, 2025.
- [15] P. K. Aithal *et al.*, "Real-time portfolio management system utilizing machine learning," *IEEE*, 2023.
- [16] H. Thanh Nhon *et al.*, "Predicting stock returns using machine learning combined with technical indicators," *PLOS ONE*, 2025.
- [17] A. Rubesam, "Machine learning portfolios with equal risk contributions," *Emerging Markets Review*, 2022.
- [18] D. Chun *et al.*, "Forecasting returns with machine learning and optimizing global portfolios," *Financial Innovation*, Springer, 2024.
- [19] M. Darwish *et al.*, "Stock market forecasting using LLMs," *arXiv*, 2025.
- [20] H. Ko *et al.*, "Portfolio optimization transformed: enhanced Black-Litterman model with AI," 2025.
- [21] J. Gao *et al.*, "Enhancing investment performance of Black-Litterman model with AI hybrid system," *Expert Systems with Applications*, 2024.
- [22] A. Chattopadhyay *et al.*, "Portfolio optimization proxies under label scarcity," *arXiv*, 2026.
- [23] "Machine learning in emerging market portfolio construction," 2024.
- [24] E. F. Fama and K. R. French, "A five-factor asset pricing model," *J. Financial Economics*, 2015.
- [25] R. Novy-Marx, "The other side of value," *J. Financial Economics*, 2013.
- [26] M. Kang, "How large-cap exclusion affects alpha and factor stability in the Korean FF3 model," 2025.
- [27] J. Traut, "What we know about the low-risk anomaly," 2023.
- [28] R. Arnott *et al.*, "Fundamental indexation," 2004.
- [29] O. A. Obrimah, "Is the Fama and French three factor model robust," 2025.
- [30] Z. Tan *et al.*, "Stock selection with random forest," 2019.
- [31] M. X. Hanauer *et al.*, "Machine learning and the cross-section of emerging market stock returns," 2023.
- [32] P. Yuan *et al.*, "Multi-factor selection and stock picking strategy," 2025.
- [33] H. Oukhouya *et al.*, "Comparing machine learning methods for forecasting the Moroccan stock market," 2023.
- [34] A. K. Sahoo *et al.*, "A hybrid deep learning model for stock price prediction," 2024.
- [35] L. Zhang *et al.*, "From factor investing to smart beta," 2024.
- [36] "Cost mitigation of factor investing," 2023.
- [37] H. Ko *et al.*, "AI enhanced Black-Litterman model," 2025.
- [38] J. Gao *et al.*, "AI hybrid system for Black-Litterman," 2024.
- [39] D. Chun *et al.*, "Machine learning portfolio optimization," 2024.
- [40] M. Darwish *et al.*, "LLM based stock forecasting," 2025.